Bank Customer Churn Prediction

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Abstract. Banks may have limited capacity to contact customers at risk of churn. We evaluate whether the model with the best overall predictive performance also identifies the most actual churners when contact is restricted to a fixed fraction of customers. On a public dataset of 10,000 bank customer records, we compare a multilayer perceptron (MLP) with Logistic Regression and Random Forest under one stratified training, validation, and test split. We report threshold-based classification metrics and rank the 1,500 held-out test customers by predicted churn probability for hypothetical contact capacities of 10% and 20%. The MLP obtains the highest F1-score (0.5980), whereas Random Forest obtains the highest ROC-AUC (0.8684). At 10% capacity, Random Forest captures 129 of 305 churners and the MLP captures 128; at 20%, the MLP captures 190 and Random Forest captures 188. These small differences show that a single overall metric does not consistently select the best contact list at every capacity. The analysis measures identification, not intervention effectiveness; uncertainty assessment remains planned.

Keywords: Customer churn · Multilayer perceptron · Contact capacity · Precision at k · Bank analytics

1 Introduction

Customer churn prediction can inform which bank customers to prioritize for retention outreach. Under a limited contact budget, a useful classifier must rank likely churners near the top, while overall accuracy and F1 alone do not specify the quality of a short contact list. Prior studies compare classical, hybrid, and neural approaches to churn [1–6], but their reported metrics rarely answer the operational question at fixed contact limits.

Research question: Does the model with the best overall classification or discrimination performance also capture the most actual churners when only the top 10% or 20% of test customers can be selected? We study (1) a controlled MLP versus Logistic Regression and Random Forest comparison on identical records, and (2) a capacity-aware evaluation using Precision@k and Recall@k. The contribution is an evaluation protocol, not a new neural architecture or observed retention intervention. The project fits ICBITS 2027 Track 6, which includes business decision support and

AI in financial services. The following sections review related work, describe the data and methods, and report preliminary results and remaining experiments.

2 Literature Review

Classical and hybrid churn methods establish the need for strong baselines. De Caigny et al. [6] combined logistic regression with tree-based segmentation and evaluated AUC and top-decile lift on customer-churn benchmarks. Their ranking measure motivates our fixed-capacity analysis; their experiments do not compare our three models on the same bank split. Zhang et al. [1] evaluated a multi-task network on the same 10,000-customer bank dataset and reported 83.61% accuracy, 0.6235 F1 and 0.8259 AUC. Its second task is derived from product count rather than measured longitudinal downsell, and these aggregate metrics do not answer which model best fills a short contact list. Their published results are informative context, not a controlled comparison with our split and pipeline.

Neural-network studies motivate explicit architecture and tuning choices. Xu et al. [2] compared TabNet, MLP and CNN on bank data with hundreds of features and reported the best overall performance for MLP (93.21% accuracy, 0.9317 F1 and 0.9245 AUC). Their data differ substantially from ours, and reported class-balance details warrant caution; numerical scores should not be compared directly. Gkonis and Tsakalos [3] studied deep-learning hyperparameters and resampling under imbalance; Domingos et al. [5] studied network hyperparameters for bank churn. Tonati et al. [4] examined counterfactual ensembles and included an MLP benchmark, illustrating that neural networks need not lead on structured churn data. These papers focus primarily on model configuration, prediction, or explanation, while we assess the ordering of customers at two contact limits.

Table 1. Selected peer-reviewed work. The reported comparisons motivate our evaluation of overall model rankings and contact lists at two capacities.

Ref.	Year	Dataset	Method	Metric/result	Relevant limitation
[1]	2026	10k bank	Multi-task network	Acc .8361; F1 .6235; AUC .8259	Derived target; no top-k
[2]	2026	Bank, 343 features	TabNet/MLP/CNN	MLP: Acc .9321; F1 .9317; AUC .9245	Different data; class details
[3]	2025	Bank churn	DNN, resampling	Tuning compared	Overall metrics only
[4]	2025	Churn datasets	Counterfactual ensemble	MLP included	Explanation focus
[5]	2021	Bank churn	Deep-network tuning	Settings compared	Network settings
[6]	2018	Churn benchmarks	Logit leaf	AUC, top-decile lift	No shared bank split

3 Dataset

We use the Churn for Bank Customers CSV from Kaggle [7], listed under CC0 Public Domain. Its 10,000 rows and 14 columns include a binary target, Exited, with 2,037 churners (20.37%) and 7,963 non-churners (79.63%). There are no missing or completely duplicate records. RowNumber, CustomerId and Surname are omitted from predictors; ten original variables remain. Six numeric variables are standardized, Gender is binary encoded, Geography is one-hot encoded into three indicators, and two already-binary variables retain their values, yielding 12 model inputs. All transforms are fitted on training records only; no augmentation or resampling is used.

Table 2. Shared stratified 70/15/15 data split (random seed 42).

Set	Records	Exited=1	Churn rate
Training	7,000	1,426	20.37%
Validation	1,500	306	20.40%
Test	1,500	305	20.33%

4 Method

4.1 Pipeline and baselines

The same seed-42 stratified 70/15/15 partition is used by all three models. The training set fits both the preprocessing transform and models; the external validation set selects configurations by ROC-AUC; and the test set is held back for the final evaluation. Within each split, rows are ordered by RowNumber. Logistic Regression selects C from {0.01, 0.1, 1, 10, 100}, with lbfgs and 2,000 maximum iterations; C=0.01 is selected. Random Forest uses 300 trees, maximum depth in {None, 8, 12}, and minimum leaf size in {1, 5, 10}; depth 8 and leaf size 1 are selected. These are finite search grids, so selection does not establish global optimum.

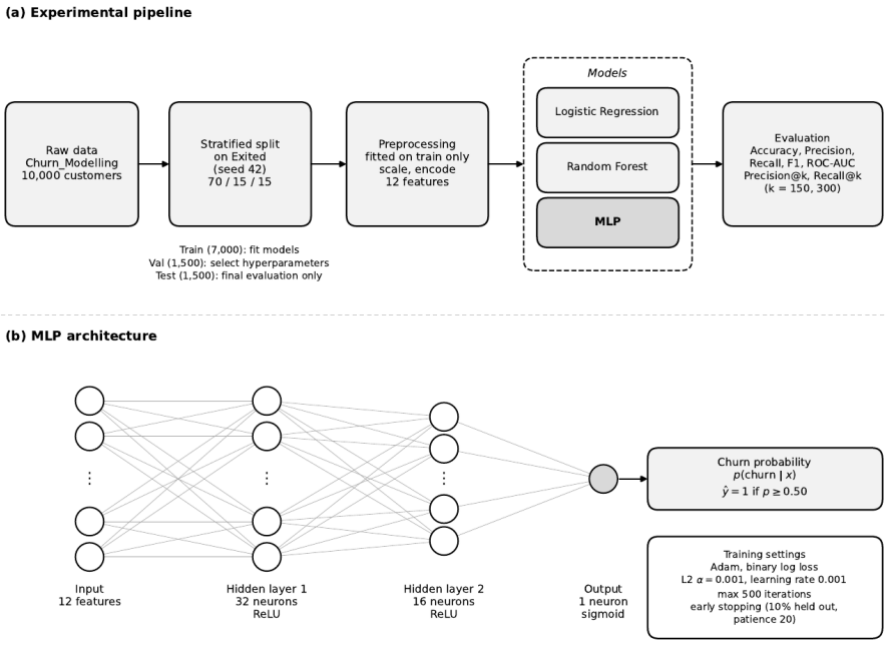


Fig. 1. Original group pipeline: shared stratified split, preprocessing fitted on training data, baselines and MLP, then evaluation.

4.2 MLP architecture and training

We compare hidden-layer widths (32, 16) and (64, 32), selecting (32, 16) by external validation ROC-AUC. For 12-dimensional input x, the selected network computes h1 = ReLU(W1 x + b1), h2 = ReLU(W2 h1 + b2), and p = sigmoid(W3 h2 + b3), where W and b are learned weights and biases and p estimates Pr(Exited=1 | x). Binary cross-entropy is minimized by Adam with learning rate 0.001 and L2 parameter alpha=0.001. The maximum is 500 iterations, with early stopping after 20 iterations without improvement on an internal 10% slice of the 7,000 training records. The external 1,500-record validation set remains separate for architecture selection. Training was conducted on CPU using Python 3.12 and scikit-learn 1.8.0 with random seed 42. Dependencies and reproduction steps are documented in the repository README and requirements file.

4.3 Evaluation and top-k proof

At threshold 0.5, we compute churn-class precision TP/(TP+FP), recall TP/(TP+FN), F1 = 2PR/(P+R), and accuracy (TP+TN)/n; ROC-AUC uses unthresholded predicted probabilities. Customers are sorted by descending churn probability, with ties resolved by ascending RowNumber and CustomerId. For k=150 and k=300, Precision@k = (actual churners in top k)/k and Recall@k = (actual churners

in top k)/305. This distinguishes classification quality from the use of a short ranked list.

Induction proof (selection by predicted score). Let $p1 \geq p2 \geq ... \geq pn$ be the sorted scores. Claim: for any k, the first k customers maximize the sum of scores over all subsets of size k. At k=1, p1 is maximal. Assume the first k maximize the sum for every size-k subset. For any size-(k+1) subset S, at least one selected element j has $j \geq k+1$; hence $pj \leq p(k+1)$. Removing j leaves a size-k subset T, whose score sum is at most $\Sigma(i=1..k)$ pi by the hypothesis. Thus $sum(S) \leq \Sigma(i=1..k+1)$ pi, attained by the first k+1. This proves the claim; ties may yield multiple optima. The property concerns predicted scores, not observed churners or actual campaign benefit.

5 Initial Results and Remaining Experiments

All three models were evaluated on the same 1,500 test customers. The MLP has the highest F1 and recall; Random Forest has the highest accuracy, precision and ROC-AUC. The primary threshold is 0.5 for all models.

Table 3. Held-out test metrics. MLP precision is 0.7550 from the saved aligned predictions.

Model	Acc.	Prec.	Recall	F1	AUC
LR	.8127	.6818	.1475	.2426	.7956
RF	.8673	.8397	.4295	.5683	.8684
MLP	.8647	.7550	.4951	.5980	.8648

Table 4. Contact-list results. RF captures one more churner than MLP at 10%, while MLP captures two more at 20%. These differences are descriptive for a single split and do not establish statistical significance. A random ranking has expected Precision@k of 20.33% and expected Recall@k of 10% or 20%, respectively.

Capacity	Model	Churners captured	Precision@k	Recall@k
10%, k=150	LR	91	.6067	.2984
	RF	129	.8600	.4230
	MLP	128	.8533	.4197
20%, k=300	LR	156	.5200	.5115
	RF	188	.6267	.6164
	MLP	190	.6333	.6230

Remaining before the final paper: paired bootstrap uncertainty intervals using common resampled customers and recomputed top-k lists; plots for Precision@k and Recall@k; explicit comparison against at least two published studies with quoted versus reproduced results distinguished; and a discussion of fairness and generalizability. There are no observed retention interventions or causal treatment effects in this dataset.

6 References

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7 Course Appendix

The project repository is available at https://github.com/DEEMALAJLAN132005/bank-churn-topk. The accompanying project archive contains the source code, dependency list, data, saved splits and predictions, and reproduction instructions. AI tools supported language editing and code integration.